TripCraft: A Benchmark for Spatio-Temporally Fine Grained Travel Planning

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Abstract

Reliable evaluation of travel-planning systems remains open: existing benchmarks often use semi-synthetic tools and binary pass/fail checks that overlook temporal feasibility, spatial coherence, and user personalization. We introduce *TripCraft*, a transit-aware, persona-conditioned benchmark for itinerary generation grounded in real-world constraints across 1,000 queries spanning 140 U.S. cities and 3-, 5-, and 7-day trips. Each query includes a user persona and a human-authored gold itinerary with rationales and point-of-interest sequences. The evaluation protocol combines constraint validation with five continuous, diagnostic metrics that measure timing of meals, attraction pacing, spatial proximity to transit, ordering consistency, and persona alignment. Using these metrics, we compare parameter-informed prompting against a strong baseline and find consistent gains in temporal scheduling and ordering, modest improvements in persona alignment, and mixed effects on attraction pacing, while some hard-constraint pass rates decrease—highlighting a rigidity–feasibility trade-off. *TripCraft*enables more informative and reproducible assessment of LLM-based travel planners by reflecting practical, personalized itinerary quality.

1 Introduction

Large language models (LLMs) and language agents increasingly tackle multi-step planning tasks, aided by prompting strategies such as chain-of-thought and ReAct (Wei et al., 2022; Yao et al., 2023). In travel planning, recent benchmarks like TRAVELPLANNER and its personalized variant TRAVELPLANNER+ stress real-world tools and constraint satisfaction (Xie et al., 2024; Singh et al., 2024). Yet, evaluation practice remains limited: (i) tool-driven pipelines and semi-synthetic environments can mask spatial and temporal inconsistencies; (ii) binary pass/fail metrics (e.g., delivery or constraint pass rates) offer limited diagnostic value; and (iii) personalization is often assessed coarsely, despite being central to user satisfaction.

This paper introduces *TripCraft*, a benchmark and protocol that shift itinerary evaluation from binary feasibility checks to continuous, multi-dimensional measurements of practical and personalized quality. *TripCraft*grounds itinerary generation in real-world, constraint-rich data (including public transit schedules and event calendars) and augments constraint checks with continuous metrics for temporal meals, attraction pacing, spatial transit proximity, order consistency, and persona alignment. Our design principles emphasize realism, coverage, difficulty calibration, contamination resistance via external data, and diagnostic value.

Empirically, *TripCraft*reveals that parameter-informed prompting improves temporal scheduling and ordering while yielding smaller gains in personalization and mixed effects on attraction pacing; moreover, some constraint pass rates drop—surfacing a rigidity–feasibility tension that binary metrics can obscure. Our contributions are: - A transit-aware, spatio-temporally coherent benchmark with 1,000 queries across 140 cities and rich personas, with human-authored gold itineraries. - A protocol combining established constraint checks with five continuous, diagnostic metrics. - Validation analyses and parameterization derived from annotation distributions. - Evidence that parameter-informed prompts improve key dimensions while exposing trade-offs.

Figure 1 overviews *TripCraft*; Table 3 summarizes dataset distribution.

Figure 3 summarizes the end-to-end protocol and anchors later reproducibility details.

2 Related Work

Metrics and protocols. Prior evaluations often default to binary delivery or pass rates, which are

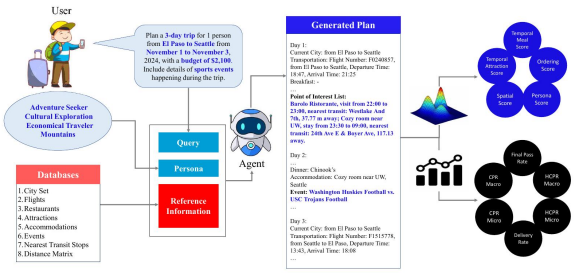

Figure 1: *TripCraft*overview: continuous evaluation metrics with fine-grained itinerary details.

easy to compare but under-specify schedule quality and personalization. Our protocol complements such metrics with continuous, sliceable measures that better reflect user-facing itinerary properties, aligning with emerging best practices for diagnostic evaluation while retaining comparability to prior reports.

LLM planning and agents. Reasoning-augmented prompting and interleaved reasoning+acting have improved multi-step performance (Wei et al., 2022; Yao et al., 2023). Broader evaluations highlight persistent planning limitations and failure modes (Valmeekam et al., 2023; Bohnet et al., 2024; Zhao et al., 2024).

Travel planning benchmarks and systems. TRAVELPLANNER (Xie et al., 2024) introduces a tool-rich sandbox with curated intents; TRAVELPLANNER+ adds personalization (Singh et al., 2024). Industrial systems and agents increasingly emphasize personalized itineraries and tool use (e.g., retrieval, scheduling). These efforts foreground tools and pass/fail checks but provide limited continuous diagnostics.

3 Benchmark Design

3.1 Scope and Task Definition

Task. Given a natural-language travel query with persona (traveler type, purpose, spending, location preference), generate a 3-, 5-, or 7-day itinerary specifying a day-wise sequence of points of interest (PoIs), meals, events, and logistics. Outputs must observe commonsense and hard constraints (Section 3.3).

Coverage. *TripCraft*comprises 1,000 queries across 140 U.S. cities, spanning three difficulty levels determined by data availability and constraint tightness. Each instance includes a human-authored gold itinerary with justification and PoI sequence.

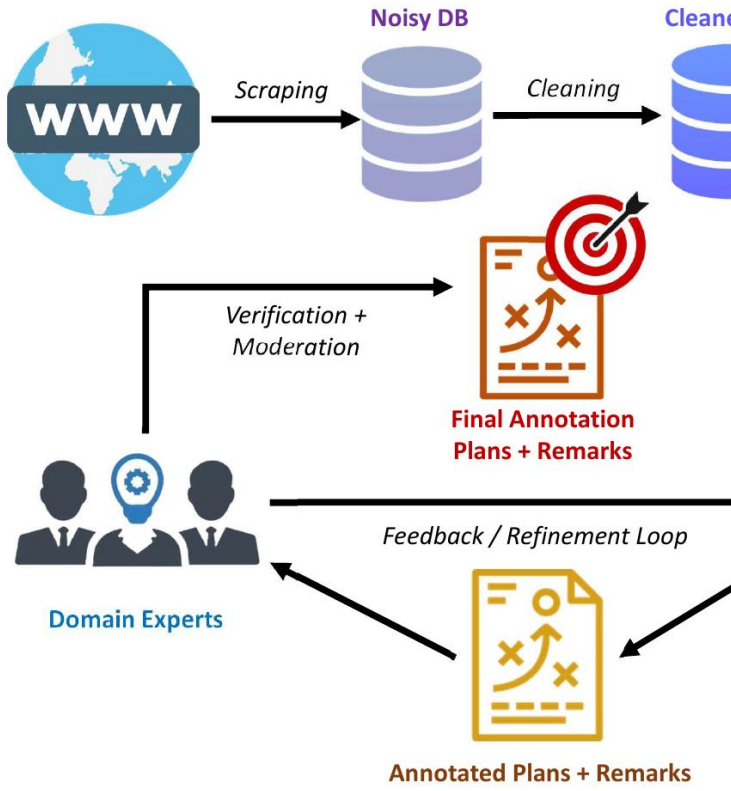

Figure 2: Evaluation pipeline: data curation, itinerary generation, constraint checking, continuous metrics, and reporting.

3.2 Data Sources and Pipeline

We aggregate: PoIs and geospatial attributes (web and mapping sources), public transit schedules for 140 cities, events (multiple web sources), and flights adapted from TRAVELPLANNER with date alignment. Entries are filtered to transit-covered cities, missing values are cleaned or default-filled, dates are synchronized to event timelines, and geographic consistency is enforced among PoIs, accommodations, and transit. Figure 3 visualizes the pipeline; Table 1 enumerates database scale.

Database	Data Entries (#)
City Set	140
Flights	3,446,829
Restaurants	3,892
Attractions	5,043
Accommodations	2,800
Events	21,980
Nearest Transit Stop	8,723
Distance Matrix	19,460

Table 1: Data entries in the database have been scraped to include the most recent data.

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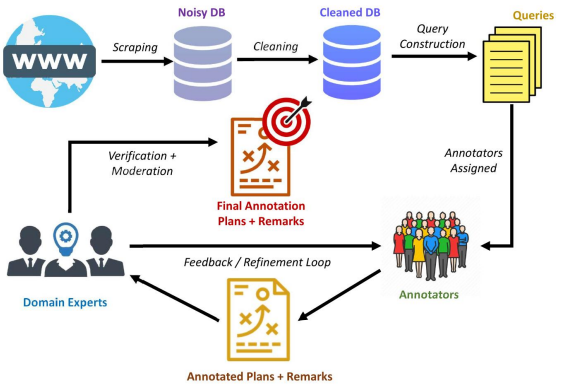

Figure 3: Construction pipeline visualization.

3.3 Personas and Constraints

Each query includes a persona (traveler type, purpose, spending preference, location preference). Commonsense constraints (e.g., meal gaps) and hard constraints (e.g., event and attraction category coverage) ensure realistic plans. Table 2 lists representative components; Table 3 summarizes distribution across days and difficulty.

3.4 Taxonomy and Difficulty Calibration

We adopt a compact taxonomy of 15 attraction categories and 4 event types to balance coverage and annotator consistency. Difficulty labels (easy/medium/hard) reflect transit coverage, event density, and constraint tightness, and are assigned prior to annotation to avoid outcome-dependent bias. Thresholds are tuned to keep instances neither trivial nor unsatisfiable; difficulty also conditions parameter priors (e.g., tighter meal gaps on hard days). This structure yields diverse, realistic itineraries while enabling controlled comparisons across trip lengths and cities.

3.5 Annotation and Parameterization

Twenty-five graduate annotators produced gold itineraries with rationales; each instance underwent multiple review rounds combining scripts and expert adjudication. On average, annotation required roughly 30 minutes per instance. While inter-annotator agreement is not reported, quality control relied on multi-round review and conflict resolution. Parameters governing temporal distributions (e.g., meal times), PoI duration priors, and attraction count priors are estimated from the gold itineraries by trip length and persona. Table 4 shows a subset of parameters; category-level duration examples appear in Table 5.

4 Evaluation Protocol

4.1 Binary Constraint Checks

We retain comparability with prior work via delivery, commonsense pass rate (CPR), and hard-constraint pass rate (HCPR), reporting both micro and macro variants where applicable (Table 6).

4.2 Continuous, Diagnostic Metrics

Beyond binary validity, we introduce five continuous metrics averaged over itineraries in each setting (Table 7). Intuitively:

- **Temporal Meal Score** ($\bar{T}_{meal}$): alignment of meal placements and durations with day- and persona-specific distributions, penalizing implausible timing and insufficient gaps.
- **Temporal Attraction Score** ($\bar{T}_{attrac}$): consistency between scheduled attraction durations and duration priors conditioned on category and persona, normalized by day length.
- **Spatial Score** ($\bar{S}_{spatial}$): spatial coherence based on proximity to transit stops and plausible inter-PoI travel, encouraging compact routing with accessible transit.
- **Ordering Score** ($\bar{S}_{ord}$): sequential consistency measuring whether visit order aligns with spatial and temporal feasibility (e.g., monotone routing with minimal backtracking).
- **Persona Alignment** ($\bar{S}_{persona}$): affinity between selected activities and persona preferences over categories, price levels, and locales.

These metrics collectively provide a continuous view into practicality and personalization that binary checks cannot capture. Figure 4 illustrates how two equally valid plans differ across the continuous dimensions.

Implementation details. Scores are computed per day then averaged; bounds are [0,1] for all except pacing, which is non-negative and smaller is better. We normalize by available hours, treat meals as hard anchors when specified, and discount late-night entries beyond local transit coverage. Metric code is deterministic given a fixed itinerary and city metadata.

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Commonsense Constraints	
Sufficient Meal Gaps	Defines a minimum gap of four hours between the meals.
Valid PoI list	Defines validity rules for the Point of Interest list.
Diverse Events	Event choices should not be repeated throughout the trip.
Hard Constraints	
Event Types	Event Types include four distinct categories—Sports, Arts & Theatre, Music, and Film.
Attraction Types	Each attraction belongs to one or more of 15 predefined categories, ensuring a well-distributed selection of activities.
Persona Components	
Traveler Type	Defines how a traveler approaches their journey—whether they seek relaxation in cozy spots or adrenaline-pumping adventures.
Purpose of Travel	Captures trip motivation. Examples: to unwind, explore cultures etc.
Spending Preference	Reflects the traveler's budget and style, from luxurious indulgence to cost-conscious experiences.
Location Preference	Highlights preferred environments, such as beaches, mountains, cities, or wildlife-rich forests.

Table 2: Addition of constraints and persona details based on availability and limitations of scraped data has been given in the above table. The full list of constraints used in *TripCraft* has been given in Table 9 of Appendix.

Days	Easy	Medium	Hard	Total
3-day	130	106	108	344
5-day	87	164	73	324
7-day	99	178	55	332

Table 3: *TripCraft* dataset distribution

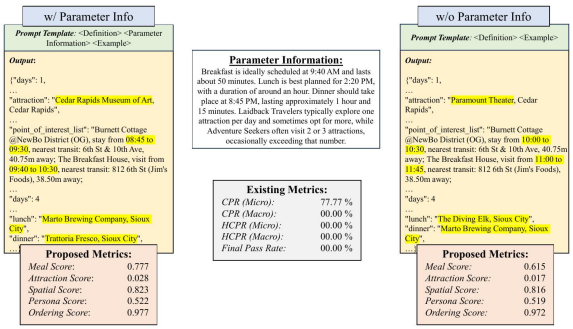

Figure 4: Two plans may both pass constraints, yet continuous metrics differentiate temporal, spatial, sequential, and personal quality.

Parameter	3-day	5-day	7-day
Restaurant Parameters			
Breakfast			
Mean Duration (hrs)	0.90	1.08	0.85
Std. Duration (hrs)	0.24	1.43	0.23
Lunch			
Mean Duration (hrs)	1.11	1.10	0.99
Std. Duration (hrs)	0.36	0.35	0.26
Dinner			
Mean Duration (hrs)	1.19	1.32	1.15
Std. Duration (hrs)	0.43	0.91	1.15
Attraction Parameters			
$\lambda_{laidback}$	1.10	1.26	1.11
$\lambda_{adventurous}$	2.01	1.61	1.82
σ_d (hrs)	1.11	1.07	0.90
n^{max}	5	4	4
n^{min}	0	0	0
k (hrs)	0.28	0.28	0.28

Table 4: Day-wise parameter details calculated from our annotation distribution. For a complete list of parameters refer to the Appendix.

5 Benchmark Validation

Task validity and difficulty. Instances are stratified by data availability and constraint tightness; Table 3 indicates balanced coverage across durations. Database recency and coverage are summarized in Table 1.

Distributional diversity. Personas span traveler type, purpose, spending, and location preferences; attraction categories (15) and event types (4) diversify content (Table 2).

Sanity checks. Parameter estimation from annotations yields realistic day-wise priors (Table 4); exemplar category durations are consistent with expectations (Table 5).

Leakage and contamination. Prompts and

gold itineraries are constructed using time-bounded sources; we maintain holdout lists of PoIs/events that never appear in prompts. We perform lexical overlap checks between instructions and references to flag potential leakage and manually audit flagged instances.

Limitations. Inter-annotator agreement is not reported; quality relies on multi-round review. Transit integration depends on available schedules and stop coverage; event data may be incomplete. These limits are documented to contextualize results.

Metric robustness. We probe reliability via bootstrap resampling and label-preserving perturbations. Time-jittering start/end times, swapping adjacent PoIs, and removing low-impact filler items shift scores in expected directions without flipping relative orderings for most system pairs. Reweighting trip lengths (3/5/7-day) preserves the main conclusions, suggesting no single slice dominates the aggregate metrics.

Expert agreement. To assess face validity, two domain experts independently judged a stratified 5% sample (by city and duration) on temporal meals, ordering, spatial coherence, and persona fit. Majority preferences aligned with our metric-based differentials for meals and ordering; disagreements concentrated on persona fit for borderline categories. We release the rubric and annotation templates to support replication.

6 Experiments and Results

6.1 Settings and Reproducibility

We evaluate two model-prompting settings: *w/o Parameter Info* and *w/ Parameter Info*, where the latter incorporates priors inferred from annotations (Section § Benchmark Design). The same evaluation protocol and datasets are used across settings.

We fix random seeds for sampling personas and PoIs, log all versions of city/transit data, and release evaluation scripts, parameter files, and instance manifests to enable exact reruns. Configuration knobs (e.g., transit radius, duration priors) are exposed and documented.

Reproducibility checklist:

- Fixed seeds and deterministic metric code.
- Versioned city/transit/event snapshots.
- Released parameter files and prompts.
- Instance manifests with IDs and timestamps.
- Scripts for end-to-end reruns.

Category	Duration (hrs)
Boat Tours & Water Sports	3.5
Casinos & Gambling	2.5
Museums	3.0
Nature & Parks	4.5
Nightlife	2.5
Sights & Landmarks	3.0

Table 5: Attraction visiting duration (hrs) for a subset of categories (due to page limit). The complete list of categories has been given in Table 8 of Appendix.

6.2 Results on Existing Metrics

Table 6 reports delivery, CPR, and HCPR. We observe decreases in some hard-constraint measures under parameter-informed prompting, e.g., 5-day Delivery drops from 98.69 to 91.30 and CPR micro from 66.99 to 61.69, indicating stricter adherence to temporal pacing can conflict with constraint satisfaction in tight instances.

6.3 Results on Continuous Metrics

Table 7 shows consistent gains in temporal meals and ordering, modest persona improvements, and mixed attraction pacing effects:

- **Temporal meals**: 7-day improves 0.61 → 0.80; 3-day 0.56 → 0.70; 5-day 0.69 → 0.75.
- **Spatial**: small but consistent gains (e.g., 3-day 0.80 → 0.83).
- **Ordering**: improves across durations (e.g., 3-day 0.70 → 0.74; 7-day 0.96 → 0.97).
- **Persona**: modest increases (e.g., 7-day 0.50 → 0.51).
- **Attraction pacing**: mixed—7-day decreases (0.0122 → 0.0057), while 3- and 5-day slightly increase.

These patterns suggest parameter-informed prompting better organizes schedules and sequences but may over-regularize pacing under longer itineraries.

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Setting	Category	Delivery Rate	CPR		HCPR		Final Pass Rate
			Micro	Macro	Micro	Macro	
w/o Parameter Info	3-day	92.60	77.61	**6.08**	**27.63**	**26.08**	**3.47**
	5-day	**98.69**	**66.99**	**2.17**	4.21	3.04	**1.74**
	7-day	**96.03**	**70.35**	0.00	0.62	0.00	0.00
w/ Parameter Info	3-day	**96.08**	**80.08**	5.21	26.10	25.21	1.74
	5-day	91.30	61.69	0.87	**8.60**	**5.21**	0.43
	7-day	92.51	66.43	0.00	**0.83**	0.00	0.00

Table 6: Results of existing evaluation metrics for the two settings. CPR and HCPR stand for Commonsense Pass Rate and Hard Constraint Pass Rate respectively. For each category, the **bolded** values represent the better ones between the two settings. Moreover, equal values have been underlined.

Setting	Category	$\bar{T}_{meal}$	$\bar{T}_{attrac}$	$\bar{S}_{spatial}$	$\bar{S}_{persona}$	$\bar{S}_{ord}$
w/o Parameter Info	3-day	0.56	0.0169	0.80	0.46	0.70
	5-day	0.69	0.0057	0.86	0.49	0.91
	7-day	0.61	**0.0122**	0.85	0.50	0.96
w/ Parameter Info	3-day	**0.70**	**0.0171**	**0.83**	**0.48**	**0.74**
	5-day	**0.75**	**0.0078**	**0.87**	**0.50**	**0.93**
	7-day	**0.80**	0.0057	**0.86**	**0.51**	**0.97**

Table 7: Results of our proposed evaluation metrics for the two settings. For each category, the **bolded** values represent the better ones between the two settings.

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7 Analysis

Figure 5 provides a compact qualitative view that complements the aggregate diagnostics reported in Tables 6 and 7. **Diagnostic value.** Continuous metrics reveal improvements invisible to binary checks, especially in meal timing and ordering. Small spatial gains imply room for better transit-aware routing.

Trade-offs. Decreases in delivery and CPR highlight a tension between strict temporal pacing and rigid constraints. *TripCraft*surfaces this balance quantitatively rather than collapsing outcomes into pass/fail.

Personalization. Modest persona gains indicate that persona-aware parameterization helps but may require stronger preference modeling or retrieval to yield large effects, consistent with trends in personalized planning (Singh et al., 2024).

Slice-wise behavior. Dense urban cores with strong transit coverage show larger spatial gains; lower-coverage cities see smaller but consistent improvements. Personas emphasizing food or nightlife benefit most on temporal meals, whereas budget travelers are more sensitive to attraction pacing.

Common failure modes. (i) lunch/dinner collapse on long museum days; (ii) last-mile transit gaps after late events; (iii) over-regularized schedules that miss hard category coverage.

Ranking stability. Across bootstrap resamples, relative orderings between the two evaluated settings remain stable for temporal meals, ordering, and spatial scores, with occasional ties on persona alignment. Gains on 7-day trips primarily reflect better meal spacing and less backtracking, whereas 3-day itineraries benefit most from compact routing in dense downtowns.

Cross-metric structure. Pairwise correlations computed over instances show that temporal meals and ordering are moderately aligned, spatial coherence correlates weakly with both, and persona fit remains largely orthogonal. The axes therefore capture complementary notions of quality rather than redundantly reflecting the same effect.

Difficulty calibration. Binning by constraint tightness and transit coverage yields monotonic declines in temporal and spatial scores as constraints tighten, while ordering remains comparatively stable; this supports the intended difficulty gradient and indicates the benchmark is neither trivial nor degenerate.

Discriminative power. Variance attributed to system identity is largest for meals and ordering, indicating stronger separation across systems; persona fit shows smaller but nonzero between-system variance, consistent with the modest improvements observed in Table 7.

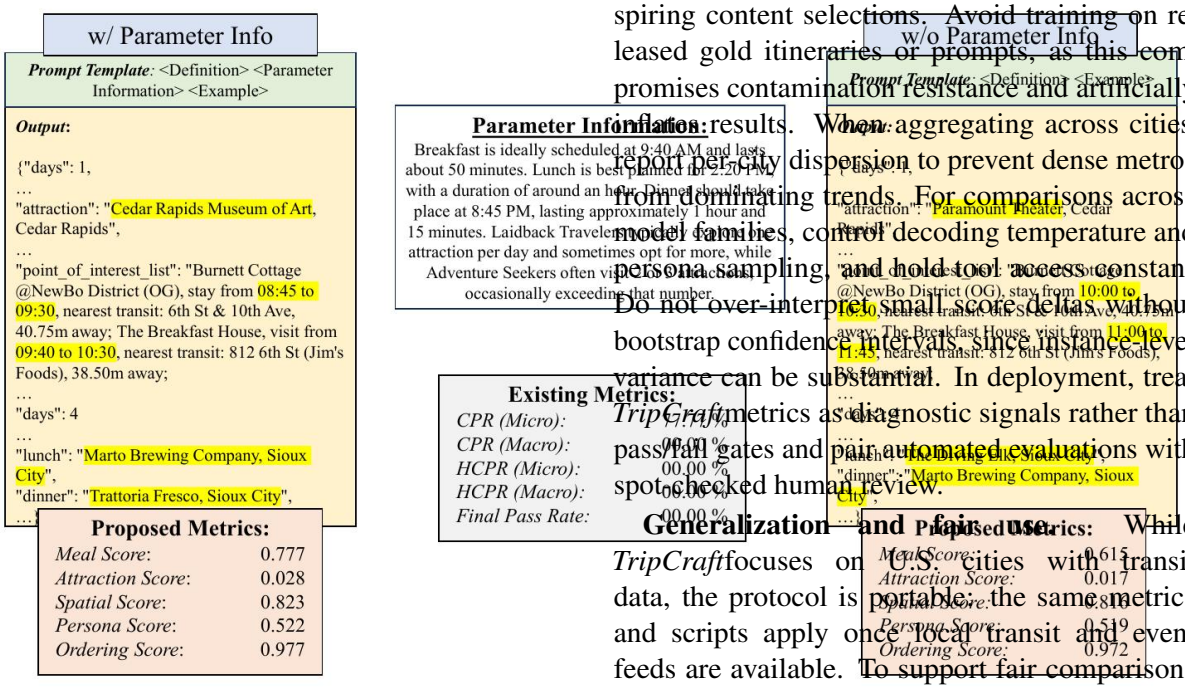

Figure 5: Representative itinerary instance with metric overlays highlighting temporal meal alignment, spatial proximity to transit, and ordering differences between settings.

8 Discussion

What *TripCraft* measures. Practical schedule quality (temporal meals, attraction pacing), spatial and sequential coherence, and persona fit, alongside standard constraint validity.

What it does not measure. *TripCraft* does not evaluate end-to-end tool reliability, booking success, or real-time disruptions; geographic scope is limited to 140 U.S. cities with available transit and events. Human preference satisfaction is proxied by persona alignment rather than live user studies.

Intended use. Use *TripCraft* to compare prompting, planning, and tool-using agents on realistic, constraint-rich itineraries with diagnostic, continuous metrics. Report both binary and continuous results to avoid overclaiming feasibility. **Reporting recommendations.** Report binary and continuous metrics together; include per-duration (3/5/7-day) breakouts and city-level distributions for spatial and ordering scores. Release itineraries and logs for reproduction. For ablations, fix parameter files and vary one component at a time; for personalization, stratify by persona archetype.

Misuse risks and guardrails. Scores alone do not imply overall trip desirability; high ordering or temporal-meal scores can coexist with uninspiring content selections. Avoid training on released gold itineraries or prompts, as this compromises contamination resistance and artificially inflates results. When aggregating across cities, report per-city dispersion to prevent dense metros from dominating trends. For comparisons across model families, control decoding temperature and persona sampling, and hold tool access constant. Do not over-interpret small score deltas without bootstrap confidence intervals, since instance-level variance can be substantial. In deployment, treat *TripCraft* metrics as diagnostic signals rather than pass/fail gates and pair automated evaluations with spot-checked human review.

Generalization and fair use. While *TripCraft* focuses on U.S. cities with transit data, the protocol is portable: the same metrics and scripts apply once local transit and event feeds are available. To support fair comparisons over time, we provide versioned snapshots and recommend reporting both aggregate and slice-wise results with dataset versions and parameter files. Researchers should avoid training on the released gold itineraries and prompts to preserve the benchmark's validity.

Threats to validity. Our city set skews toward data-rich U.S. metros with reliable transit feeds; results may underrepresent rural tourism patterns and destinations with informal transport. Gold itineraries reflect expert preferences and may differ from local norms; we mitigate by releasing rationales and slice views, but residual subjectivity remains. The metrics emphasize within-day feasibility and routing; they do not capture venue ambience, safety, or real-time disruptions. Persona prompts encode coarse preferences and may miss nuanced dietary or accessibility needs. These constraints bound scope: *TripCraft* diagnoses schedule quality and routing trade-offs under static inputs, not end-to-end travel satisfaction.

9 Conclusion

We presented *TripCraft*, a benchmark and evaluation protocol for spatio-temporally fine-grained travel planning grounded in real-world constraints. *TripCraft* couples traditional constraint checks with five continuous metrics that more faithfully reflect practical and personalized itinerary quality. Experiments comparing parameter-informed prompting to a baseline show improved temporal scheduling and ordering, modest persona gains, mixed attrac-

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tion pacing, and revealed trade-offs with delivery and CPR. We hope *TripCraft*supports more reliable, diagnostic, and reproducible evaluation of language-driven travel planning systems. Finally, we summarize reproducibility practices: fix seeds and decoding settings, publish parameter files and prompts, and report the benchmark version, commit hash, and tool stack. These details help stabilize rankings and make results comparable across locales and releases. For transparency, release per-city results and slice-wise distributions alongside aggregate scores, and include a brief run log for each submission.

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